



Work and Energy M1

Formal Lab report

Introduction to Engineering & Practices
FE1073

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Abstract

This report examines the principle of conservation of mechanical energy with the aid of a compound pendulum.

This is where your introduction or main paragraph text goes. Because we used the `\section*{}` command with an asterisk, LaTeX will not automatically number this section, keeping your lab report looking clean and custom-formatted.

You can write your regular data analysis here. If you need to include a quick formula, it's as simple as writing $E = mc^2$ inline, or giving it its own dedicated line like this:

$$f(x) = \frac{\Delta y}{\Delta x} \tag{1}$$

Acknowledgements

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1 Introduction

The law is called the conservation of energy. It states that there is a certain quantity, which we call energy that does not change in the manifold changes which nature undergoes.

Richard Feynman

The principle of conservation of energy states that, when a system and its surroundings undergo a transition from an initial state to a final state, the change in energy is zero [1]. In this context, the total mechanical energy of our closed system should remain constant, as it is converted between its gravitational potential energy, as well as its kinetic forms. This can be better illustrated with the following equation.

$$\Delta E = \Delta E_{\text{system}} + \Delta E_{\text{surroundings}} = 0. \quad (1)$$

This experiment investigates this principle with the use of a compound pendulum.

A compound pendulum refers to a rigid body that is extended and free to rotate about a fixed horizontal axis, without passing through its centre of mass. It greatly differs from an ideal simple pendulum: A simple pendulum is able to idealise the system with a point mass on a string, as the string's mass is relatively low and thus negligible relative to the bob. It is ignored in theoretical calculations. A simple pendulum setup can be seen in the diagram below.

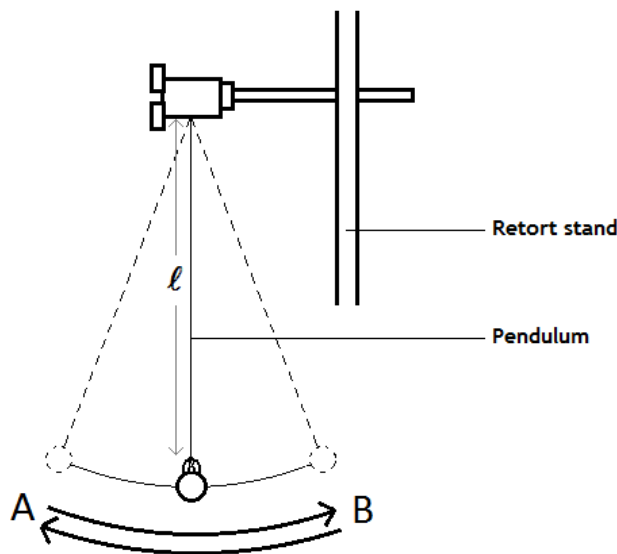


Figure 1: A simple pendulum apparatus setup [2].

However, this configuration is susceptible to experimental inaccuracies if the string loses tension during oscillation. Conversely, a compound pendulum overcomes this limitation by having a rigid body. It has its mass distributed throughout the entire body, leading to it having a different centre of mass as an ideal system, as well as a moment of inertia about the pivot, compared to an ideal simple pendulum. this experimental setup can be seen in Figure 3. For this experiment, to minimise frictional losses, our pendulum is suspended on a knife-edge pivot, then released from rest at a known angular displacement, and allowed to oscillate under gravity.

1.1 Experimental Objective

The intended outcomes of this experiment are as follows:

- a. demonstrate conservation of mechanical energy in the swinging of a pendulum
- b. understand work done by gravitational force,
- c. understand potential energy in a gravitational field,
- d. rotational kinetic energy of a solid body,
- e. prove the conservation of energy in a closed system, and
- f. moment of inertia about an axis.

1.2 Theoretical Framework

Consider an object near the earth's surface, subject to its gravitational field. Within its closed system, it is held at a height, increasing its potential energy. When released, the gravitational force, acting upon it as a net external force, does a positive amount of work, with kinetic energy increasing until the object reaches its lowest point at the earth's surface, where theoretically, the force acting on it should now be zero. Conversely, if the object is given an upwards velocity, a negative amount of work will be done onto the object by gravitational force, with its kinetic energy reaching 0, while its gravitational potential energy reaches its highest.

If we view it in the context of a pendulum, with the rigid body for our setup holding the pendulum at a pivot, our kinetic energy should be fully recovered at its highest point of the swing. The following energies can be best represented with the following equations and variables.

Kinetic Energy K :

$$K = \frac{1}{2}I\left(\frac{d\theta}{dt}\right)^2 \quad (2)$$

where I represents moment of Inertia of the pendulum about the pivot.

1.2.1 Exercise 1

The angular velocity of the pendulum ω can be represented as:

$$\omega = \frac{\text{Angular Displacement}}{\text{Time}} = \frac{\lambda}{t} \quad (3)$$

velocity v refers to the rate of displacement, which can be thought as displacement s over time.

$$v = \frac{ds}{dt} \quad (4)$$

$$s = L_g\theta \quad (5)$$

where L_g is the distance between centre of mass of pendulum and pivot.

$$\frac{ds}{dt} = \frac{L_g\theta}{dt} \quad (6)$$

Kinetic Energy K can thus be represented as:

$$K = \frac{1}{2}M\left(\frac{ds}{dt}\right)^2 = \frac{1}{2}ML_g^2\left(\frac{d\theta}{dt}\right)^2 \quad (7)$$

M being the mass of the pendulum.

1.2.2 Exercise 2

If we compare equations (2) and (7), we can thus infer that I can be expressed as the following:

$$I = ML_g^2 \quad (8)$$

The formula for L_g in terms of mass M and period T for small oscillations of the pendulum can be seen by making a direct comparison of the pendulum and the linear vibration of a spring-mass system, as shown in Figure 2.

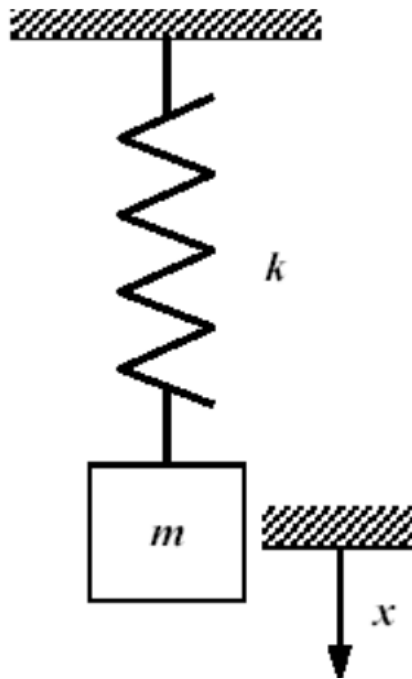


Figure 2: Linear spring-mass system [3].

The kinetic energy of the vibrating mass can be calculated as:

$$K = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 \quad (9)$$

where mass is given as m , displacement as x .

By Hooke's law, the spring force can be demonstrated as:

$$F_{spring} = -kx \quad (10)$$

k being the spring constant (i.e. stiffness coefficient).

1.2.3 Exercise 3

Considering that for small oscillations, moment about the pivot is approximate to the spring force, and bearing in mind that $\sin \theta \approx \theta$ for when θ is small.

$$F_{\text{pendulum}} = -Mg \sin \theta \approx -Mg\theta \quad (11)$$

The component $Mg \sin \theta$ acts perpendicular to the string, so its moment about the pivot is a restoring torque:

$$\tau = -MgL_g \sin \theta \approx -MgL_g\theta \quad (12)$$

Comparing with $F_{\text{spring}} = -kx$, where $x \leftrightarrow \theta$, $m \leftrightarrow I$ and force $\leftrightarrow$ torque:

$$k \leftrightarrow MgL_g$$

$$k = \frac{Mg}{L_g} \quad (13)$$

1.2.4 Exercise 4

From Newton's law of motion, it can be shown that the period T_{spring} can be thought as

$$T_{\text{spring}} = 2\pi\sqrt{\frac{m}{k}} \quad (14)$$

If we substitute $m \rightarrow I$ and $k \rightarrow MgL_g$ into Eq. (14), and use $I = ML_g^2$ from Eq. (8), we get:

$$T = 2\pi\sqrt{\frac{I}{MgL_g}} = 2\pi\sqrt{\frac{L_g}{g}} \quad (15)$$

1.2.5 Exercise 5

The distance L_g can be expressed as:

$$L_g = g\left(\frac{T}{2\pi}\right)^2 = \frac{gT^2}{4\pi^2} \quad (16)$$

$$L_g = \frac{9.81 \times 1.07^2}{4\pi^2} = 0.28449m \approx 28.4cm/0.284m$$

Period T of pendulum for 'small' oscillations based off our calculations:

$$T = \frac{\text{Time for 10 oscillations}}{10} = \frac{10.7}{10} \approx 1.07s \text{ (3.s.f.)}$$

Mass M of pendulum = 0.228kg, Theoretical L_g = 0.284m, Measured L_g = 0.288m

$$I = ML_g^2 = 0.228(0.288)^2 \approx 0.0189kg \text{ m}^2 \text{ (3.s.f.)}$$

Aside from calculations, the other method we could use in determining the position of the center of mass of the pendulum, involves balancing the compound pendulum over a triangular block. With the aid of a metal ruler, that was how we were able to determine the respective L_g for our experiment.

1.2.6 Exercise 6

The vertical rise of the mass from equilibrium at angle θ is:

$$h = L_g - L_g \cos \theta = L_g(1 - \cos \theta)$$

At the highest point, $\theta = \Theta$:

$$\text{P.E.} = MgL_g(1 - \cos \Theta) \quad (17)$$

$$\text{K.E.} = \frac{1}{2}I\omega^2 = \frac{1}{2}ML_g^2\omega^2$$

$$\% \text{ discrepancy} = \frac{\text{P.E.} - \text{K.E.}}{\text{P.E.}} \times 100\%$$

Plotting K.E. (y-axis) against P.E. (x-axis) should give a straight line through the origin. If energy is conserved, the **gradient is 1**, since Eq. (1) says total energy within a closed system should be conserved. A gradient slightly below 1 indicates energy losses from friction at the pivot and air resistance. This answers Discussion (4).

Applying Newton's second law, with $s = L\theta$:

$$ML\ddot{\theta} = -Mg \sin \theta$$

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0$$

For small oscillations ($\theta < 5^\circ$, so $\sin \theta \approx \theta$):

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \quad \Rightarrow \quad \omega_n = \sqrt{\frac{g}{L}}$$

$$T = 2\pi\sqrt{\frac{L}{g}} \quad (18)$$

Potential Energy U gained at the highest point:

$$U = mgh \quad \Rightarrow \quad h = L_g(1 - \cos \theta) \quad \Rightarrow \quad U = \mathbf{mgL_g}(1 - \cos \theta) \quad (19)$$

With $L = L_g$, this is identical to $T = 2\pi\sqrt{L_g/g}$. This is expected, as it is similar to the pendulum with a point mass on a light string.

Conservation of energy between the highest and lowest points gives:

$$\frac{1}{2}ML^2\omega^2 = MgL_g(1 - \cos \Theta) \quad (20)$$

The mass cancels:

$$\omega = \sqrt{\frac{2g(1 - \cos \Theta)}{L}} = 2\sqrt{\frac{g}{L}} \sin \frac{\Theta}{2} \quad (21)$$

The corresponding linear speed of the mass is $v = L\omega = \sqrt{2gL(1 - \cos \Theta)}$.

2 Experimental Methodology

2.1 Experimental Apparatus

- A compound pendulum
- 2 Support Stands
- 1 stopwatch
- 1 digital photo timer

2.2 Experimental Set-Up

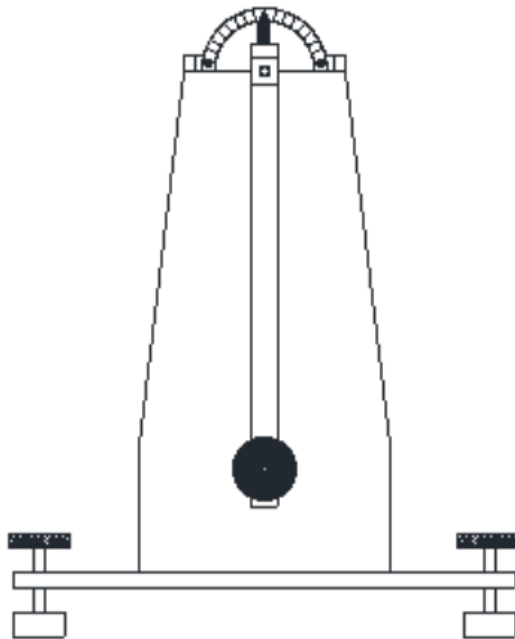


Figure 3: Experimental Apparatus set up. [3].

- Assemble the pendulum system, which consists of a long bar with a round disc at one end, a bearing pivot block equipped with a protractor, and support stands, as shown in Figure 3.
- Adjust the height of the digital photo timer sensor so that it points directly at the centre of the pendulum disc.
- Identify and note the offset reading on the protractor when the pendulum is resting in its vertical equilibrium position.

2.3 Experimental Procedure

1. Switch on the digital photo timer.
2. Manually displace the pendulum to an initial angle of $\Theta = 20^\circ$ from its established equilibrium position.
3. Ready your stopwatch, and release the pendulum.
4. Record the time (t seconds) displayed on the timer as the pendulum passes through the lowest equilibrium point for the very first time. If this initial pass is missed, the trial must be repeated.
5. Repeat steps 2 through 5 for other initial angular displacements (Θ) of 30° , 40° , 50° , and 60° .
6. To find the small angle period, do the following. Switch off the digital photo timer.
7. Displace the pendulum by a small angle of less than 5° from its equilibrium position.
8. Then, release the pendulum from rest, verifying again that the swing remains on a vertical plane.
9. Using a manual stopwatch, measure the total time taken for exactly 10 complete oscillations. This data is used to calculate the period (T) of the pendulum for small oscillations.

3 Results

3.1 Raw Data

Table 1: Raw experimental data

Trial	Θ (rad)	time (s)
1	$\frac{\pi}{9}$	0.084
2	$\frac{\pi}{6}$	0.060
3	$\frac{2\pi}{9}$	0.041
4	$\frac{5\pi}{18}$	0.034
5	$\frac{\pi}{3}$	0.028

Time period T for 10 oscillations: $10 \times T = \mathbf{10.7s}$

$\Rightarrow$ Period T of oscillation = $\mathbf{1.07s}$

Mass M of pendulum = $\mathbf{0.228kg}$

Initial pendulum offset from the 90° mark = $\mathbf{0}$

Total angular displacement (Δ) of pendulum for which timer is activated = $\mathbf{0.1396\ rad}$

Measured distance L_g of centre of gravity of pendulum from pivot line = $\mathbf{0.288m}$

Moment of Inertia I :

$$I = ML_g^2 \quad \Rightarrow \quad I = 0.228kg \times (0.288m)^2$$

$$\mathbf{I = 0.0189kg\ m^2}$$

Using the following equations,

$$\omega = \frac{\text{Angular Displacement}}{\text{Time}} = \frac{\lambda}{t}$$

$$U = mgL_g(1 - \cos \theta) \quad K = \frac{1}{2}I\omega^2$$

and % discrepancy refers to

$$\frac{PE - KE}{PE} \times 100\%$$

Table 2: Calculated data

ω (rad/s)	P.E. (J)	K.E. (J)	%discrepancy
1.66	0.0388	0.0260	33.0%
2.33	0.0862	0.0513	40.5%
3.41	0.1506	0.0110	27.0%
4.11	0.2299	0.0160	30.4%
4.99	0.3218	0.0235	27.0%

Note: P.E. refers to potential energy at the highest point of the pendulum, K.E. refers to the kinetic energy at the lowest point of the pendulum. (i.e. both of the energies are at their peaks).

3.2 Graphical Analysis

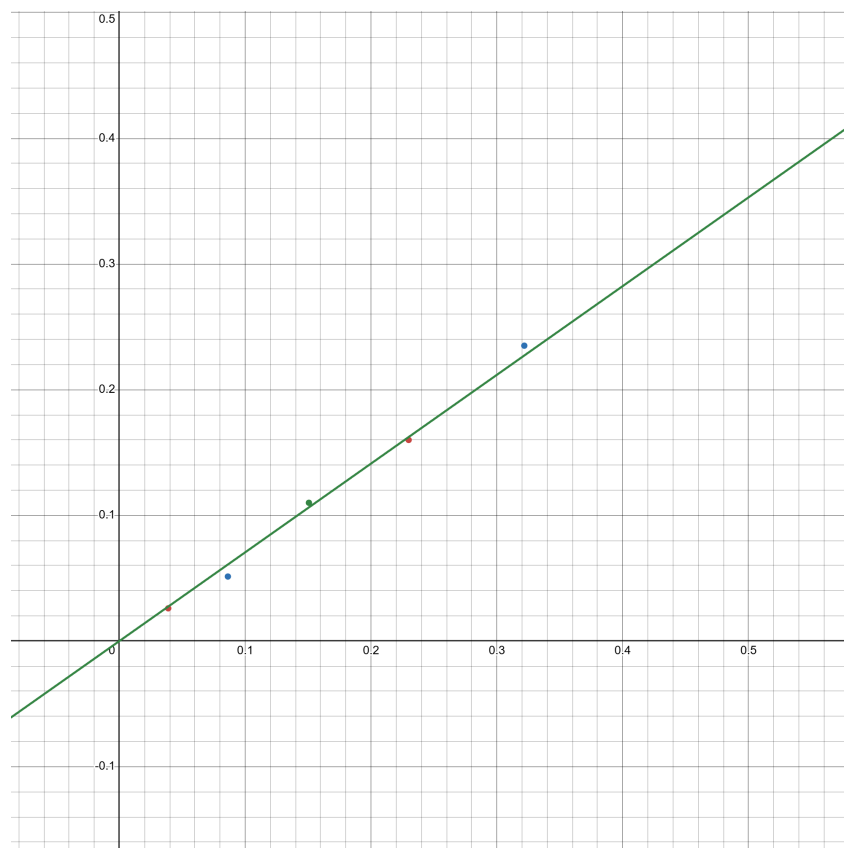


Figure 4: Graph of Kinetic Energy (J) to Potential Energy (J)

4 Discussion

4.1 Comparison of Experimental and Theoretical Data

4.1.1 7.2 (b)

Net restoring force = $-mg \sin \theta$

For small θ , $\sin \theta \approx \theta$

Net restoring force = $-mg\theta$

Since $F_{\text{net}} = ma$

$$-mg\theta = ma$$

$$-mg\theta = m\alpha L$$

$$\alpha = -\frac{g}{L}\theta$$

SHM $\rightarrow \alpha = -\omega^2\theta$

$$-\omega^2\theta = -\frac{g}{L}\theta$$

$$\omega = \sqrt{\frac{g}{L}}$$

$$\frac{2\pi}{T} = \sqrt{\frac{g}{L}}$$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

From exercise 4,

$$T = 2\pi\sqrt{\frac{I}{MgL_g}}$$

Equating the two expressions for T (as indicated by the curly bracket):

$$\frac{I}{MgL_g} = \frac{L}{g}$$

$$I = ML_gL$$

$$ML_g^2 = ML_gL$$

$$L_g = L$$

Thus, Compound pendulum can be treated as a simple pendulum if we know the distance between CG to pivot.

4.1.2 7.3 (c)

$$\text{KE of pendulum} = \frac{1}{2} I \left(\frac{d\theta}{dt} \right)^2$$

$$\text{PE gained at highest point} = mgL_g(1 - \cos \theta)$$

$$\text{Since } m = M, L_g = L,$$

$$\text{PE gained at highest point} = \text{KE at lowest point}$$

$$\begin{aligned} MgL(1 - \cos \theta) &= \frac{1}{2} I \left(\frac{d\theta}{dt} \right)^2 \\ MgL(1 - \cos \theta) &= \frac{1}{2} ML^2 \omega^2 \\ \omega^2 &= \frac{2g}{L} (1 - \cos \theta) \\ \omega &= \sqrt{\frac{2g(1 - \cos \theta)}{L}} \end{aligned}$$

(a) What is a conserved and non-conserved force? Give an example of conserved and non-conserved force.

A conserved force refers to where the total work done in moving an object between two points is entirely independent of the path taken. If an object moves in a closed loop and returns to its exact starting position, the net work done by a conservative force is zero, meaning mechanical energy is preserved within the system. Examples will include Gravitational Force, and Electrostatic Force.

A non-conserved force refers to a force when its total work done, depends on the path taken by the object, which causes mechanical energy to be changed from the system, through thermal energy, or soundwaves. Examples of this includes kinetic friction and air resistance.

(c) Determine the center of mass of the compound pendulum through experiment and determine the distance, L_g , between the center of mass of the compound pendulum and the pivot. Compare with the value evaluated according to Eq. (9), is it reasonable to simplify the compound pendulum as a simple pendulum?

The pendulum had a mass M of 0.228 kg. Ten small-angle oscillations took 10.7 s, giving a period T of 1.07 s. The distance from the pivot to the centre of gravity was taken as $L_g = 0.288$ m, based off our measured values, which agrees to within about 1% with the value of 0.28449 m given by Eq. (16), $L_g = \frac{gT^2}{4\pi^2}$. The moment of inertia about the pivot was therefore $I = ML_g^2 = 0.0189 \text{ kg m}^2$. From protractor readings of 87° to 95° , the angular displacement over which the photo timer was active was $\lambda = 0.1396 \text{ rad}$ (8°). The angular velocity at the lowest point was obtained from $\omega = \frac{\lambda}{t}$, the kinetic energy from K.E. $= \frac{1}{2} I \omega^2$, and the potential energy from P.E. $= MgL_g(1 - \cos \Theta)$. The results are summarised in Table 2.

Both energies increased with release angle, as expected. In every trial the kinetic energy at the lowest point was smaller than the potential energy at the highest point. The discrepancy ranged from 27.0% to 40.5%, with a mean of 31.6%.

(b) What is the gradient of the graph of K.E. against P.E.? What can you conclude from this?

As observed from Figure 4, our graph of K.E. against P.E. gave a best fit straight line passing through the origin. Its gradient, taken from two points on the line well separated from the data, (0.13, 0.09) J and (0.30, 0.21) J, was

$$y = mx + c$$

where m stands for the gradient, and c the y -intercept. For our graph, we took 2 plot points and calculated that

$$m = \frac{0.21 - 0.09}{0.30 - 0.13}$$

$$m = 0.70588 \approx 0.706$$

$$c = 0; \text{ since graph passes through the origin}$$

The point at 30° lies furthest below the line, consistent with its being the largest discrepancy in Table 1.

4.2 Error Analysis

(d) What are the possible sources of error for the experiment? Give estimates for the percentage errors involved.

The dominant uncertainty is likely λ . Since $\omega \propto \lambda$ and K.E. $\propto \omega^2$, any error in λ is doubled in the kinetic energy. Assuming a protractor reading uncertainty of $\pm 0.5^\circ$ at each end of the detection window, λ carries an uncertainty of about $\pm 1^\circ$ in 8° , or about 12.5%, which corresponds to roughly 25% in K.E. This is comparable to the observed discrepancy.

As a consistency check, we have also redone this experiment at least 3 times before we recorded our readings, and we noted our compound pendulum has a greater tendency to swing to its right, further agreeing with our findings that the dominant uncertainty is λ . Another check we did would be comparing the values. If energy were exactly conserved, the measured times would imply λ values of 0.163–0.181 rad, with a mean of 0.169 rad (about 9.7°). This is only about 1.7° larger than the value used, within the reading resolution of the protractor.

The data therefore cannot distinguish genuine energy dissipation from a systematic underestimate of λ , although the latter is likely the main contributor.

Other sources of error would include the following, with percentage estimates:

- Timer resolution (± 0.001 s): about 1.2% in t at 20° and 3.6% at 60° , or up to about 7% in K.E.
- Release angle ($\pm 0.5^\circ$): about 5% in P.E. at 20° , falling to about 1.5% at 60° .
- Period (± 0.2 s stopwatch reaction over 10.7 s): about 1.9% in T , which becomes about 3.7% in L_g since $L_g \propto T^2$.
- Averaging effect: $\omega = \frac{\lambda}{t}$ is the mean angular velocity across the sensor window, which is slightly lower than the peak value at the lowest point.
- Release technique and alignment: hand release may impart a small initial velocity, and any out-of-plane swing wastes energy and contributes to friction.
- Compound versus simple pendulum: The value of L_g from Eq. (9), 0.285 m, agrees with the value of 0.288 m used in the calculations to within about 1%, which is within measurement uncertainty. It is therefore reasonable to treat the pendulum as a simple pendulum of length L_g with $I = ML_g^2$, since most of the mass is concentrated in the disc at the end of the bar. However, while it can be treated within reasonable bounds to be similar, one still cannot ignore the likely impact the distribution of weight that will have on its centre of mass, thus affecting the readings.
- Inherent friction at the pivot and air resistance: While minor, but it can still attribute up to 1 % source of error.

4.3 Suggested Improvements

Several methods we can employ to improve our readings and reduce the possible sources of errors mentioned above will include:

- To minimise human error, we can replace our eyes with a digital angle sensor. If that is really not possible, the best we could do is try to avoid parallax error, by reading protractors at eye level.
- Repeat the experiment with even more readings, and then consider taking the average from those readings to arrive at a better conclusion.
- Automate the human parts with machines. i.e. consider using a system with an integrated data logger and timer, that it automatically cuts the timing when the pendulum swings past. As of right now, the stopwatch has up to a 0.1ms resolution, however this is subject to human reaction delay times, which can add approximately at least 100 to 200ms, which proves to be fatal, considering that a 1ms resolution already contributes 3.6% error.
- By automating, we can also standardise the release, and able to repeat the experiments more consistently. Although if that's not possible, we can try to employ other means to better standardise the release, such as using electromagnets, as opposed to releasing by hand. This eliminates any potential push, or out-of-plane motion due to accidental forces exerted.
- We could consider improving the period measurements as well, where instead of timing 10 oscillations, we time about 20-30 instead, thereby reducing potential reaction-time error, affecting L_g .

5 Conclusion

The kinetic energy at the lowest point of the swing was found to be directly proportional to the potential energy at the highest point, with a gradient of 0.706 rather than the ideal value of 1. The proportionality supports the principle of conservation of mechanical energy, but the 29

References

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A Raw Data Sheets

Attach or include raw data tables here if they are too large for the main text.

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